BIOL 312 · Lecture 9 · Marine Adaptation

How corals survive a warming ocean

The cellular biology of thermal tolerance — and why some reefs are beating the heat.

Where we are going in the next fifty minutes.

01 The puzzle — reefs that should be dead, but aren't

02 The mechanism — symbiosis under heat stress

03 The evidence — the Lizard Shoal transplant study

04 The frontier — what we still cannot explain

A reef that bleached in 2016 came back greener than its neighbors.

Most of the central Coral Sea bleached when temperatures held 2 degrees above the summer mean for six weeks. Yet one shoal recovered in eighteen months while reefs forty kilometres away never did. That gap is today's question.

The textbook says corals cannot adapt this fast — the textbook is incomplete.

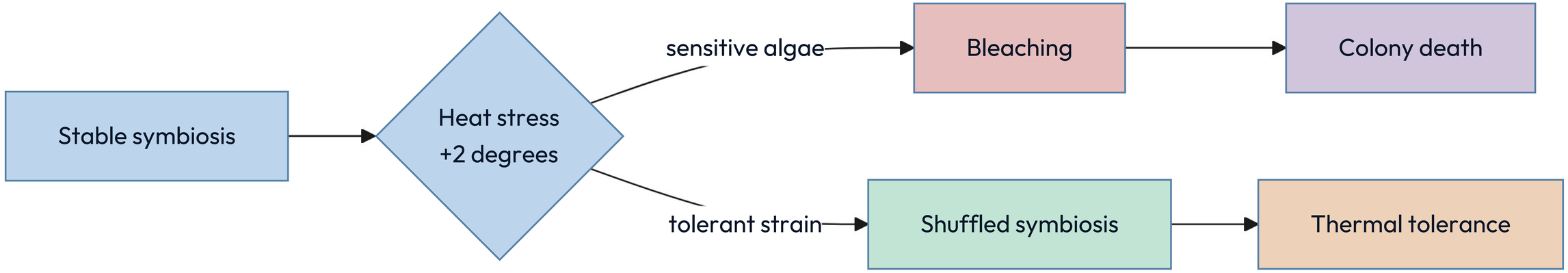
Reef-building corals are slow: a colony may take a decade to mature and centuries to build structure. Classical adaptation across that timescale cannot track a climate shifting in years. Something faster is doing the work.

- Generation time is far too long for selection alone to explain.
- The recovery happened within a single coral generation.

The coral is not one organism — it is a partnership under negotiation.

Every coral hosts millions of single-celled algae, *Symbiodinium*, that photosynthesise inside its tissue and feed it. Heat ruptures that partnership: the algae turn toxic, the coral expels them, and the white skeleton shows through. That is bleaching.

Heat tips a stable partnership into collapse — or into a swap.



The survivors did not evolve — they switched partners.

The recovered shoal hosted a different algal clade, *Durusdinium trenchii*, which tolerates heat the common *Cladocopium* strains cannot. The coral genome barely changed. The symbiont community did. Adaptation happened at the level of the partnership, not the host.

Lizard Shoal transplant · 2016–2019

What the symbiont swap bought, measured against unswapped controls.

Reciprocal transplant, n = 240 colonies, three reef sites, 36-month follow-up.

+1.5°C
BLEACHING THRESHOLD RAISED

84%
SURVIVAL VS 31% CONTROL

18 mo
TO FULL RECOVERY

−22%
GROWTH RATE TRADE-OFF

Tolerance is not free — the fast algae feed the coral less.

Colonies hosting the heat-tolerant clade grew 22% slower in normal years. The partnership that survives a heatwave is a worse provider the rest of the time. This is the central tension of thermal tolerance: resilience bought with growth.

What a coral needs to survive the next marine heatwave.

01

A tolerant symbiont in reach

The heat-adapted algal clade must already be present in the local water column.

02

Flexible host physiology

The coral must be able to host more than one symbiont strain — not all species can.

03

A recovery window

Temperatures must fall back below threshold long enough to rebuild tissue.

04

Larval connectivity

Tolerant recruits must reach the reef to reseed what heat killed.

Symbiont shuffling is real, but it is not a universal escape hatch.

Only about a quarter of reef-building genera can host multiple clades. For the rest, the partnership is fixed at settlement and cannot be renegotiated when the water warms. The flexible species may inherit the reef.

- Branching *Acropora* shuffle readily; massive *Porites* far less.
- A reef's future may hinge on which genera dominate it today.

The frontier: can we seed tolerance before the heat arrives?

Assisted evolution trials are now inoculating coral larvae with heat-tolerant clades in the lab, then settling them on degraded reefs. Early results are promising and ethically fraught — we would be engineering the partnership that nature negotiates.

- First field outplants reached the central reef in 2025.
- Whether the tolerance persists across generations is unknown.

Next lecture · larval dispersal and reef connectivity

The coral that survives changes its partner, not its genes.